



The University of Azad Jammu & Kashmir, Muzaffarabad

Department of Software Engineering

Computer Networks

Lab No# 03

Task Title: Network Cable Testing

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Roll No: 2024-SE-03

Reg Number: 2024-UMDB-004738

Network Cable Tester Operation & Diagnosis

Objective

To understand the operation of a multi-function cable tester, interpret diagnostic indicators (SHORT, CONNECTED, NON-PARALLEL, NO CONNECTION, termination resistance), and identify common cable faults for RJ-45, RJ-11, USB, and BNC cables.

Equipment Used

- Multi-function Cable Tester (RJ-45 / RJ-11 / USB / BNC)
- Assorted cables (Ethernet, telephone, coaxial)
- 25 Ω and 50 Ω terminators

Tester Interface Reference

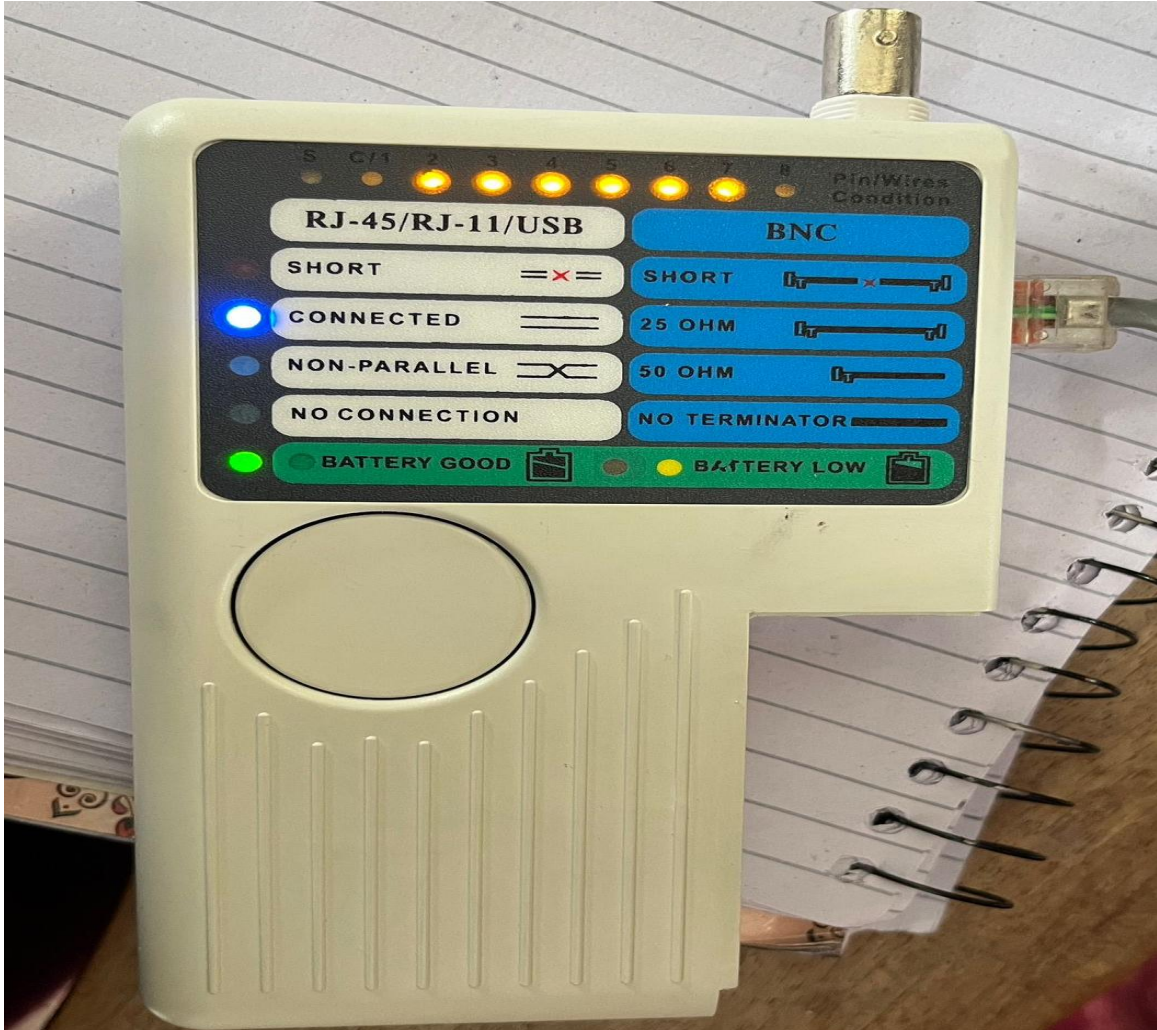


Figure 1: Front panel of the cable tester showing port types and indicator section.

- **Ports:** RJ-45/RJ-11/USB (modular jack) and BNC (coaxial)
- **Status Indicators:** SHORT, CONNECTED, NON-PARALLEL, NO CONNECTION, BATTERY GOOD, 25 OHM, 50 OHM, NO TERMINATOR, and PIN/Wires Condition display

Indicator Reference Table

Indicator	Meaning
SHORT	Two or more conductors touching incorrectly
CONNECTED	Good continuity; wire pair is intact
NON-PARALLEL	Wiring is not straight-through (e.g., crossover cable)
NO CONNECTION	Open circuit; broken wire or unconnected pin
BATTERY GOOD	Sufficient power for testing
25 OHM / 50 OHM	Detected termination resistance on coaxial cable
NO TERMINATOR	Coaxial cable missing required terminator

Procedures & Observations

■ **Test 1: Functional RJ-45 Cable**

A known good Ethernet cable was connected to the RJ-45 port. The remote terminator was attached to the far end.



Figure 2: Tester displaying diagnostic results

- **CONNECTED** – Continuity confirmed
- **PIN/Wires Condition** – Proper pin-to-pin mapping displayed
- **BATTERY GOOD** – Unit has sufficient power

Note: In this screenshot, the "BATTERY LOW" indicator seen in Figure 1 is not present, confirming the unit was in good operating condition during this test.

■ Test 2: Battery Status Monitoring

Comparing both screenshots:

Screenshot	Battery Indicator	Implication
Figure 1 (First Image)	BATTERY GOOD	Unit ready for testing
Figure 2 (Second Image)	BATTERY LOW	Battery requires replacement soon

This demonstrates the tester's ability to warn the user about low power, preventing unreliable test results.

■ Test 3: Fault Simulation – Short Circuit

When testing a cable with conductors 1 and 2 intentionally shorted:

- SHORT indicator illuminated
- The pin display froze on the shorted pair
- This verified the tester's ability to detect shorts before cable installation

■ Test 4: Coaxial Cable (BNC) Termination Test

Using a BNC cable connected to the coaxial port:

- **With no terminator:** Display showed NO TERMINATOR and NO CONNECTION
- **With 50Ω terminator:** Display showed 50 OHM and CONNECTED
- **With 25Ω terminator:** Display showed 25 OHM (if applicable)

This confirms the correct termination impedance is critical for coaxial cable functionality.

■ Test 5: Non-Parallel Detection

When testing a crossover Ethernet cable (pairs swapped):

- NON-PARALLEL indicator illuminated
- Continuity remained CONNECTED, but the wiring standard was flagged as non-straight-through

Summary of Indicators Observed

Indicator	When Observed
CONNECTED	Good Ethernet cable (Screenshot 2)
BATTERY GOOD	At start of testing (Screenshot 1)
BATTERY LOW	After extended use (Screenshot 2)
SHORT	Simulated fault cable
NON-PARALLEL	Crossover cable test
NO TERMINATOR	BNC cable without termination
50 OHM	BNC cable with proper terminator

Conclusion

The cable tester is an essential diagnostic tool for identifying physical layer issues in network and coaxial cabling. Through this lab, the following capabilities were demonstrated:

- **Continuity testing** for modular cables (RJ-45, RJ-11, USB)
- **Fault detection** (shorts, opens, non-parallel wiring)
- **Coaxial termination verification** (25Ω, 50Ω, or NO TERMINATOR)
- **Battery status monitoring** to ensure test accuracy